



Reading Data from Databases





READ ROWS



```
import pyodbc
```

```
connectionString = r'DRIVER={ODBC Driver 13  
for SQL  
Server};SERVER=.\SQLEXPRESS;DATABASE=northw  
ind;Trusted_Connection=yes'
```

```
sqlStr = " SELECT * FROM customers"
```

```
conn = pyodbc.connect(connectionString)  
cur = conn.cursor()  
result = cur.execute(sqlStr).fetchall()  
conn.close()
```

```
for row in result:  
    print(row)
```



READ ROWS WITH A FILTER



```
import pyodbc
connectionString = r'DRIVER={ODBC Driver 13 for
SQL
Server};SERVER=.\SQLEXPRESS;DATABASE=northwind;T
rusted_Connection=yes'
```

```
sqlStr = "SELECT * FROM customers WHERE
city='London'"
```

```
conn = pyodbc.connect(connectionString)
cur = conn.cursor()
result = cur.execute(sqlStr).fetchall()
conn.close()
```

```
for row in result:
    print(row)
```



READ ROWS SORTED

```
import pyodbc
connectionString = r'DRIVER={ODBC Driver 13 for SQL
Server};SERVER=.\SQLEXPRESS;DATABASE=northwind;Trusted
_Connection=yes'
```

```
sqlStr = "SELECT * FROM customers ORDER BY
contactName"
```

```
conn = pyodbc.connect(connectionString)
cur = conn.cursor()
result = cur.execute(sqlStr).fetchall()
conn.close()
```

```
for row in result:
    print(row)
```





READ ROWS - SORTED AND FILTERED



```
import pyodbc
connectionString = r'DRIVER={ODBC Driver 13 for
SQL
Server};SERVER=.\SQLEXPRESS;DATABASE=northwind;Tru
sted_Connection=yes'
```

```
sqlStr = """SELECT * FROM company
WHERE county='Devon'
ORDER BY company_no"""
```

```
conn = pyodbc.connect(connectionString)
cur = conn.cursor()
result = cur.execute(sqlStr).fetchall()
conn.close()
```

```
for row in result:
    print(row)
```



BETTER USE FUNCTIONS



```
import pyodbc

connectionString = r'DRIVER={ODBC Driver 13 for
SQL
Server};SERVER=.\SQLEXPRESS;DATABASE=northwind;Tru
sted_Connection=yes'

def readData(sql):
    conn = pyodbc.connect(connectionString)
    cur = conn.cursor()
    result = cur.execute(sql).fetchall()
    conn.close()
    return result

#-----Main code -----
res = readData("SELECT * FROM company WHERE
county='Devon' ")
for row in res:
    print(row)
```



Any questions?



Thank you!

Hope you enjoyed this learning journey!